Practice Problems
Calculus Physics II

Name: Key **Date:** _____

Chapter 57 Intro to Optics

1) Optics is the branch of physics that studies

A) Lens.
B) The human eye.
C) EM waves.
D) Glasses.
E) Bending of light.

2) An extended light source can best be described as

A) A source that has a significant size compared to the optical system
B) A source that emits light from a single point
C) A source that extends to infinity
D) A source that can be turned on or off

3) When light is absorbed by a material how what is the energy in the light transformed into?

A) Diffuse waves
B) Reflected waves
C) Diffracted waves
D) EM waves
E) Thermal energy

4) The type of light/material interaction that emits light in different directions than the incident light in small particles is called

A) Reflection
B) Scattering
C) Diffraction
D) Refraction
E) Absorption

5) Under which condition will EM waves not interact with the object?

A) When the object is much smaller than the wavelength of the EM wave.
B) When the object is much larger than the wavelength of the EM wave.
C) When the object is approximately equal to the wavelength of the EM wave.
D) EM waves will always interact with an object.

6) The reflection off a smooth surface is

A) Specular Reflection
B) Diffuse reflection

7) Diffuse reflection occurs when the size of surface irregularities is

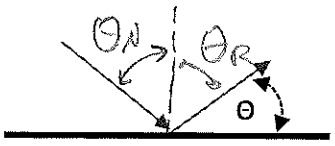
A) small compared to the wavelength of the light used.
B) large compared to the wavelength of the light used.
C) microscopic.

8) A ray of light reflects off the surface shown, it the angle measured from the surface of the reflected beam is Θ = 40°, what is the angle of reflection?

normal line

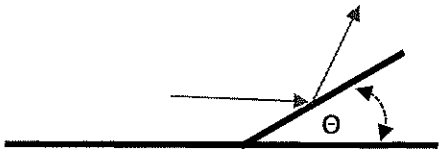
$\Theta_R = 90° - 40° = 50°$

$\Theta_i = \Theta_R = 50°$



9) A ray of light comes in parallel to the horizonal surface and reflects off a second surface that makes an angle of Θ = 30° to the horizontal surface, what is the angle of reflection of the ray off the second surface?

$\Theta_i = 90° - \Theta = 60°$



10) It is difficult to see the roadway when driving on a rainy night mainly because

A) light scatters from raindrops and reduces the amount of light reaching your eyes.
B) of additional condensation on the inner surface of the windshield.
C) the film of water on your windshield provides an additional reflecting surface.
D) the film of water on the roadway makes the road less diffuse.
E) none of the above

11) The amount of light reflected from the front surface of a common windowpane is about

A) 4%.
B) 8%.
C) 40%.
D) 92%.
E) 96%.

12) Light travels from one place to another along a path of least

A) distance.
B) time.
C) effort.
D) expense.
E) complication.

13) Refraction occurs when light passing from one medium to another and

A) changes frequency.
B) changes speed.
C) loses energy.
D) reflects inward.

14) When EM waves are absorbed by an object what happens to that object

A) Gets cooler
B) Gets warmer
C) Turns black
D) Turns white

15) When sunlight illuminates this homework page it

A) reflects most of it.
B) absorbs at the text lettering.
C) both of these
D) neither of these

16) If white light is incident on a piece of paper and the colors that are absorbed are Red, Orange, Green, Blue, Indigo and Violet, what color does the paper appear?

A) All the color that are absorbed mixed.
B) Black
C) Yellow
D) White

17) Which type of scattering of light does not depend on the frequency of light?

A) Rayleigh Scattering
B) Mie Scattering
C) They both do not depend on the frequency of light.

18) Which scattering is the scattering due to the wavelength of the incident light being approximately the same size as the particles?

A) Rayleigh Scattering
B) Mie Scattering
C) They both do not depend on the frequency of light.

19) The sky is blue because

A) The red/yellow/orange/green portions of the spectrum is absorbed by the particles in the atmosphere.
B) The red/yellow/orange/green portions of the spectrum is scattered must by the particles in the atmosphere.
C) The blue portion of the spectrum is absorbed by the particles in the atmosphere.
(D) The blue portion of the spectrum is scattered must by the particles in the atmosphere.

20) A total shadow is called

(A) an umbra, and a partial shadow a penumbra.
B) a penumbra, and a partial shadow an umbra.
C) neither of these

KEY

1) C
2) A
3) E
4) B
5) A
6) A
7) B
8) 50°
9) 60°
10) D
11) A
12) B
13) B
14) B
15) C
16) C
17) B
18) A
19) D
20) A